Synaptic and Dendritic Integration

The neuron's input–output function

Computational Neuroscience

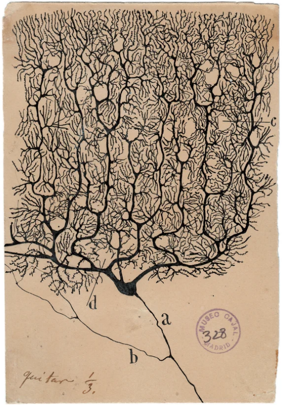
University of Bristol

Slides adapted from C O'Donnel

M Rule

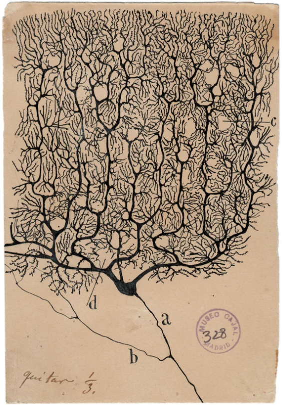
Learning outcomes:

- Describe input summation in the synapse and dendrite
- Explain how input location on the dendrites affects summation
- What are some functions of dendritic nonlinearities?



Cajal

What goes on here?



Cajal

What goes on here?

Nonlinear synaptic integration

- Many spike trains in $\rightarrow$ one spike train out
- "Point" neuron models (e.g. LIF) assume soma performs a weighted linear sum of synaptic currents.

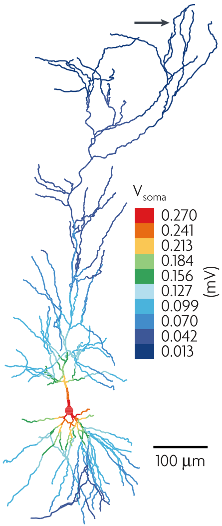
Real neurons are different!

1. Dendrites $\Rightarrow$ *spatial layout of synaptic inputs.*
2. Dendrites have voltage-dependent (active) ion channels which makes *synaptic integration nonlinear.*

Synaptic location matters

On the right is a trace of the dendritic tree of a CA1 pyramidal cell from a rat.

- Colour: amplitude of the voltage response (EPSP) at soma when the synapse is placed at the corresponding location on the dendritic tree.
- Without boosting, a synapse would give a smaller somatic response if it was located at a distal dendritic site.
- Voltage-dependent ion channels in dendrites can boost synaptic inputs to amplify their effect at the soma.

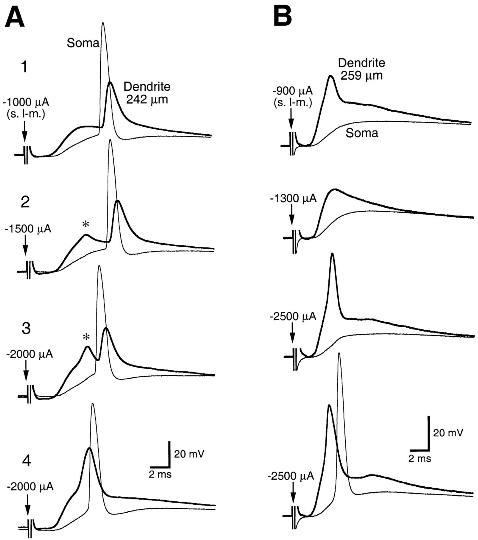


Spruston, Nat Rev Neurosci (2008)

Dendritic spikes

Some neurons have enough voltage-gated ion channels in their dendrites to enable dendritic action potentials.

- Dendritic spikes tend to be weaker, less all-or-none that axonal action potentials
- (note variable dendritic amplitudes in each plot on right).
- One dendritic spike is not always sufficient to trigger an axonal spike.

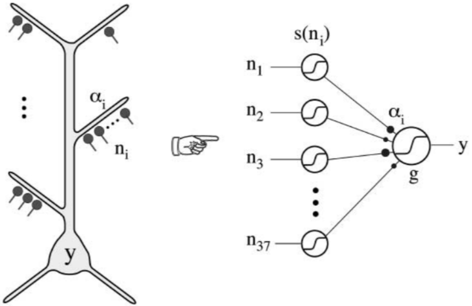


Golding and Spruston, Neuron (1998)

The single neuron as two-layer neural network

Dendritic spikes ⇒

- One pyramidal neuron ≈ multi-layer neural network.
- Each dendritic does a nonlinear operation on its inputs before passing the signal to the soma.
- Voltage-gated ion channels expand the brain's computational power.



Poirazi and Mel, Neuron (2003)

Summary

(EARLIER SLIDE DECKS)

- Ion channels are protein complexes that let electrical current flow in and out of the neuron.
- They mediate neurons' electrical dynamics.
- Kinetic models of neurons are commonly described using ODEs.
- Ion channels' voltage dependence makes the neuron's input-output function highly nonlinear.
- Single neurons can in principle approximate a 2-layer artificial neural network

end

page left blank for notes

page left blank for notes

… Quantum Chemistry, Molecular dynamics

Physiological, Quantitative

Biological Realism, Data needed to identify parameters

Molecules
Gillespie, Master Equation

Concentrations
Mass-Action Kinetics

Conductance Models
Hodgkin–Huxley

Spiking Models
Leaky Integrate and Fire

Rate Neurons
Neural Mass/Field Models

Poisson Neurons
Generalized Linear Models

Binary Neurons
McCulloch–Pitts, Hopfield, Perceptron

Computational Efficiency, Mathematical Tractability

Phenomenological, Qualitative

Cognitive Neuroscience, Psychology …